

Intelligent Mutex Execution Environment

Complete Application User Guide

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Getting Started with IMEE

Complete Application User Guide

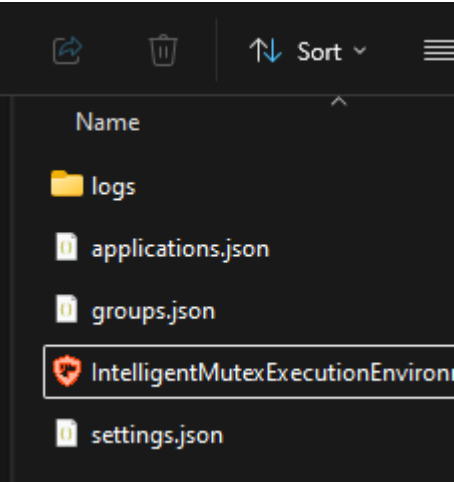
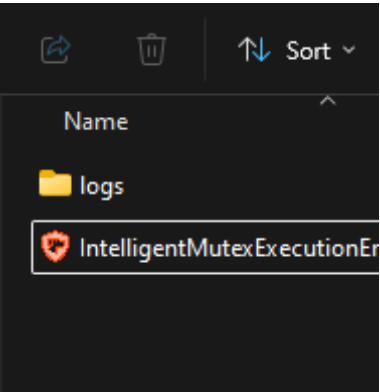
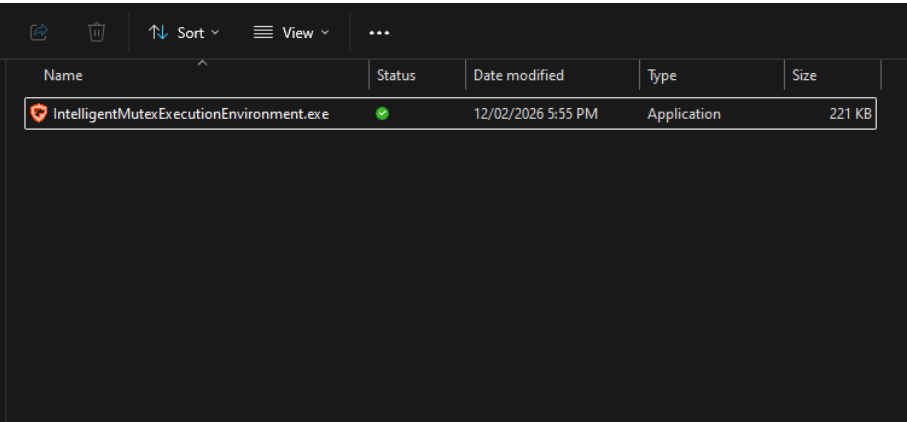
Tagline

IMEE: **I** **M**ust **E**xecute **E**xclusively

What a mutex *does* – it enforces exclusive execution, ensuring only one thread or process runs a critical section at a time.

Getting Started

IMEE Installation

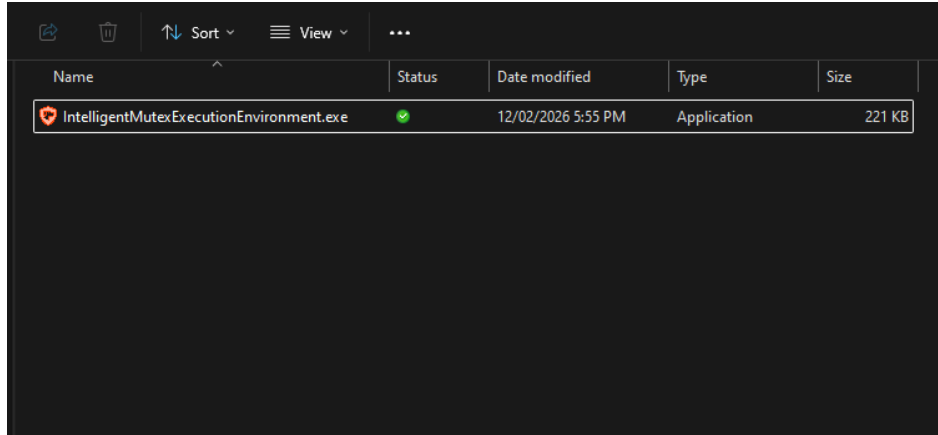


No installation is required. Copy `IntelligentMutexExecutionEnvironment.exe` to any folder and run it. The application creates its data files automatically in the same directory:

File	Purpose
application.json	Stores your managed applications
groups.json	Stores your groups
settings.json	Stores your station name, performance, and logging configuration
logs/	Daily log files

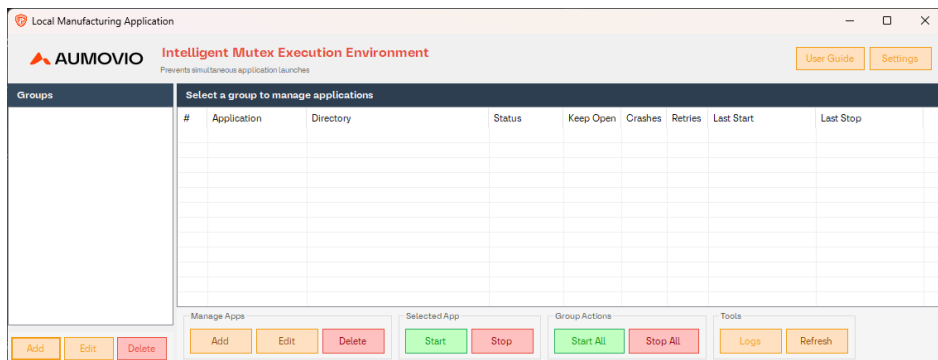
Getting Started

First launch and Startup Behavior



First launch

- Double-click `IntelligentMutexExecutionEnvironment.exe`
- The window appears in the **bottom-right corner** of your screen
 - Both the group list (left) and application list (right) will be empty



Startup Behavior

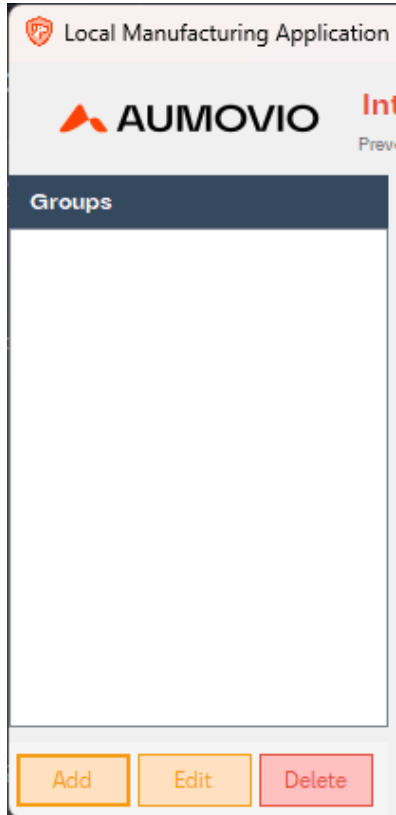
When Instance Manager starts, it checks if any managed applications are already running. If so, they are ****automatically terminated**** and a notification lists the affected apps. This ensures all managed applications are only launched through Intelligent Mutex Execution Environment.

Interface Overview for IMEE

Complete Application User Guide

Interface Overview

The interface is divided into two main areas



Left Panel - Groups

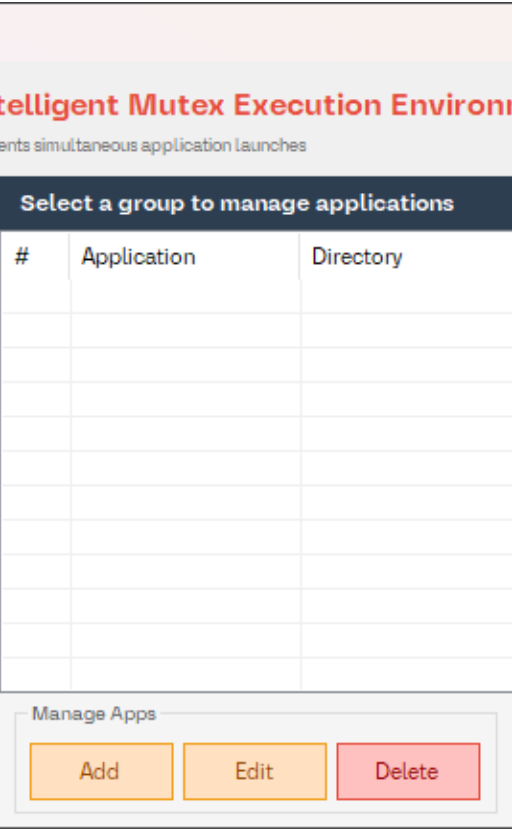
Group list : Shows all your application groups

Add Group / Edit Group / Delete Group: To Manage groups

Column	Description
Indicator	Current state (see Group Status indicators)
Group Name	Name of the Executable
N of T	Current state (see Group Status indicators)

Interface Overview

The interface is divided into two main areas



Right Panel Applications

Application List: Shows apps in the selected group with 9 columns

Action Buttons: Add, Edit, Delete, Start, Stop, Start All, Stop All, Refresh

Settings / Logs: Configure station name, performance, and logging settings, or open the logs folder

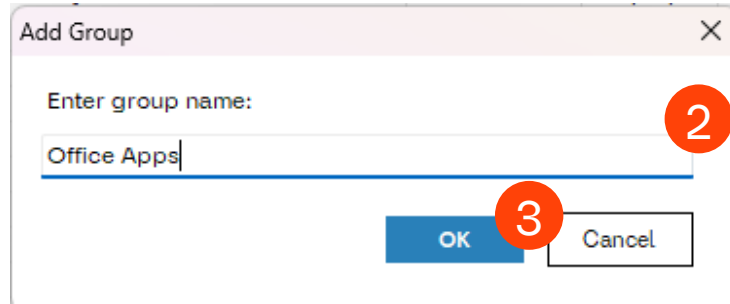
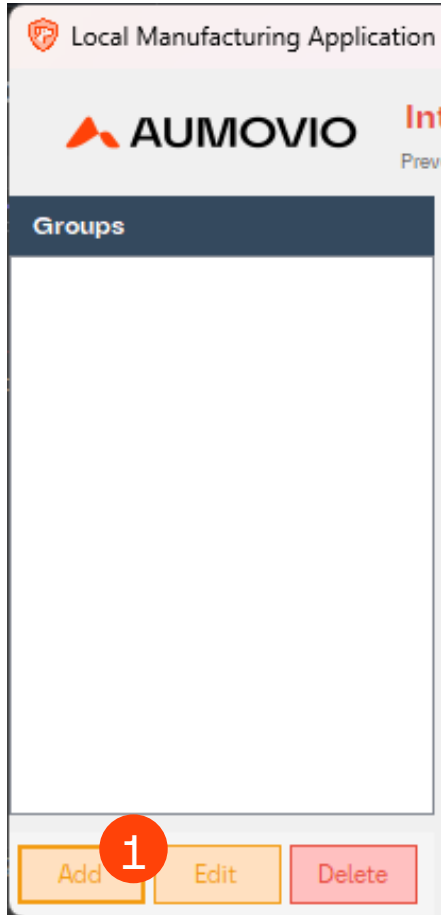
Column	Description
Index	Unique auto-incremented ID
Application	Name of the Executable
Directory	Full file path
Status	Current state (see Status indicators)
Keep Open	“Yes” if auto-restart is enabled
Crashes	Total Number of detected crashes
Retries	Current auto-restart retry count
Last Start	When the app was last started
Last Stop	When the app was last stopped

Managing Groups for IMEE

Complete Application User Guide

Managing Groups

Groups let you organize your applications logically



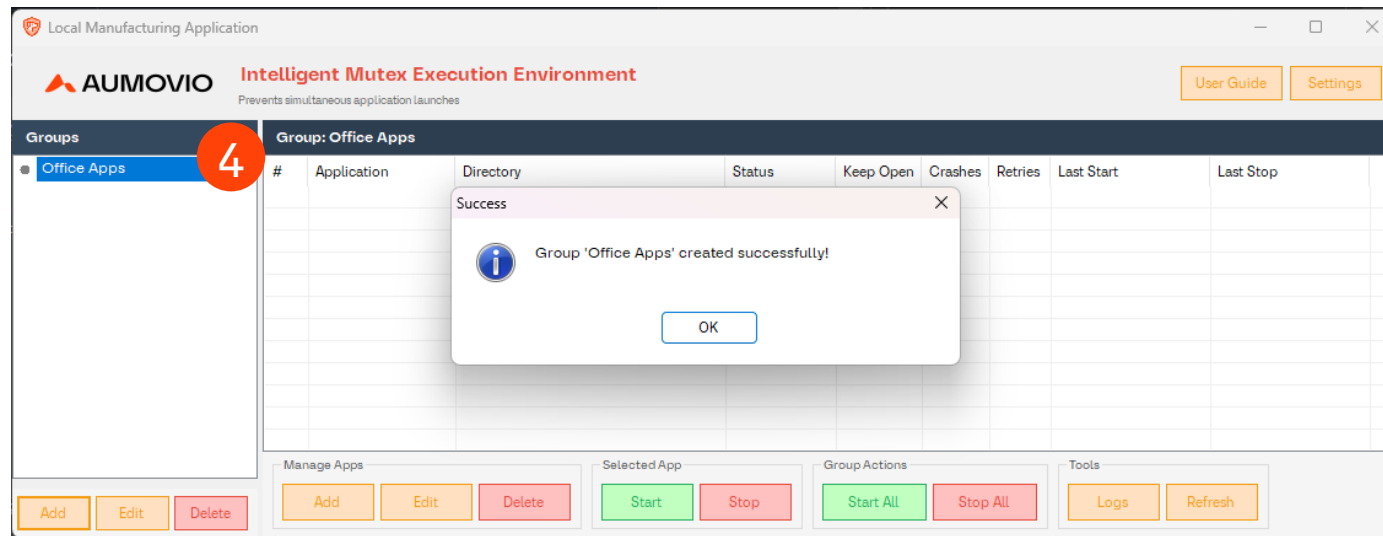
Create a Group

Step 1: Click **Add Group**

Step 2: Enter a name (e.g “CCU Top”, “ADAS SRR TOP”)

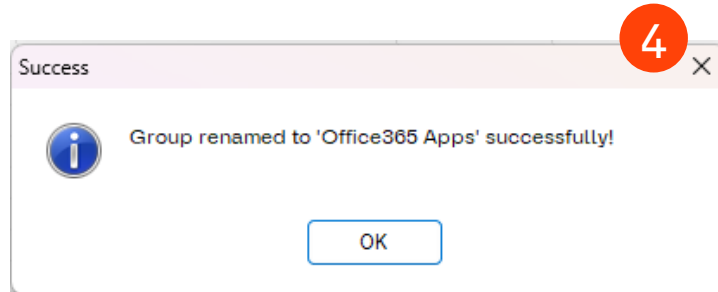
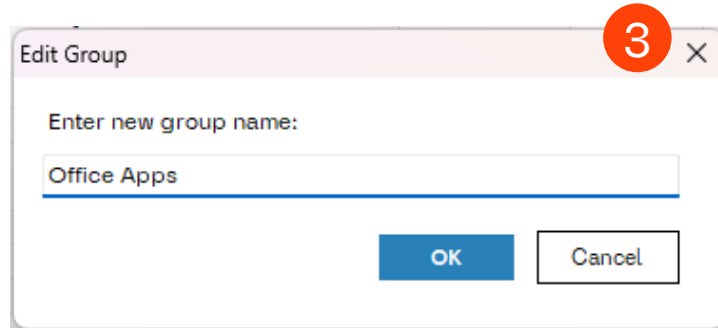
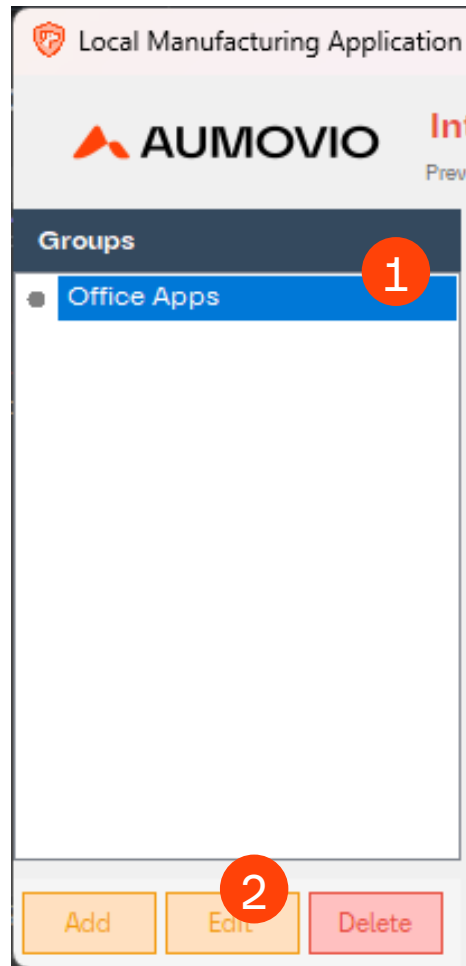
Step 3: Click OK

Step 4: The new group is created and automatically selected



Managing Groups

Groups let you organize your applications logically



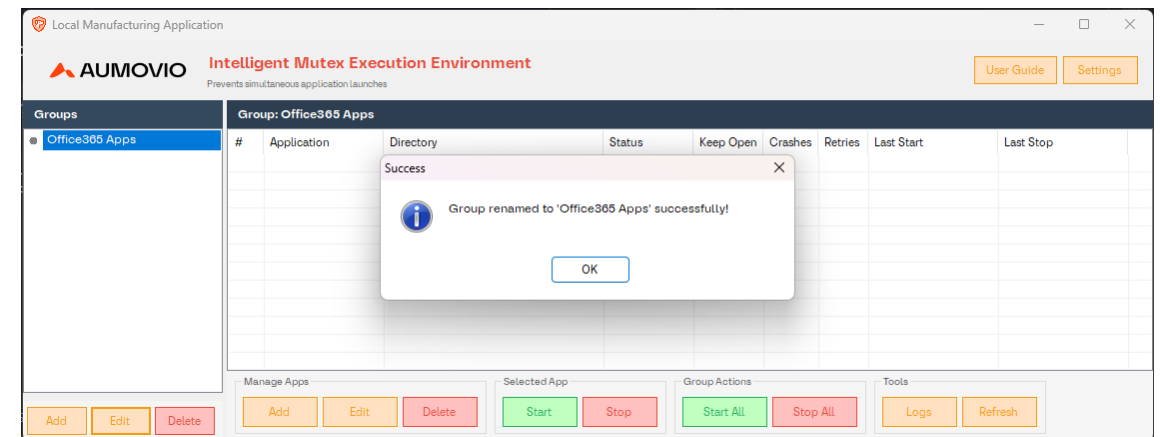
Rename a Group

Step 1: Select the group in the left panel

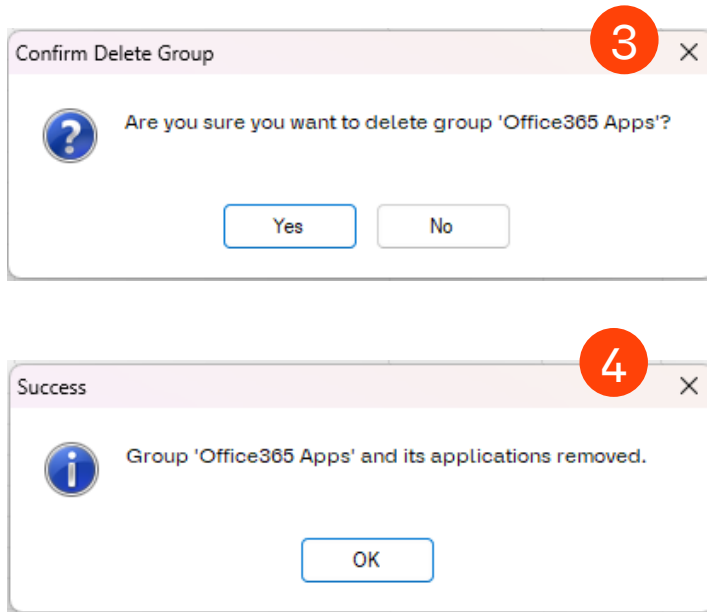
Step 2: Click Edit

Step 3: Enter the new name

Step 4: Click Ok



Groups let you organize your applications logically



Delete a Group

Step 2: Click Delete

Step 4: Click Ok

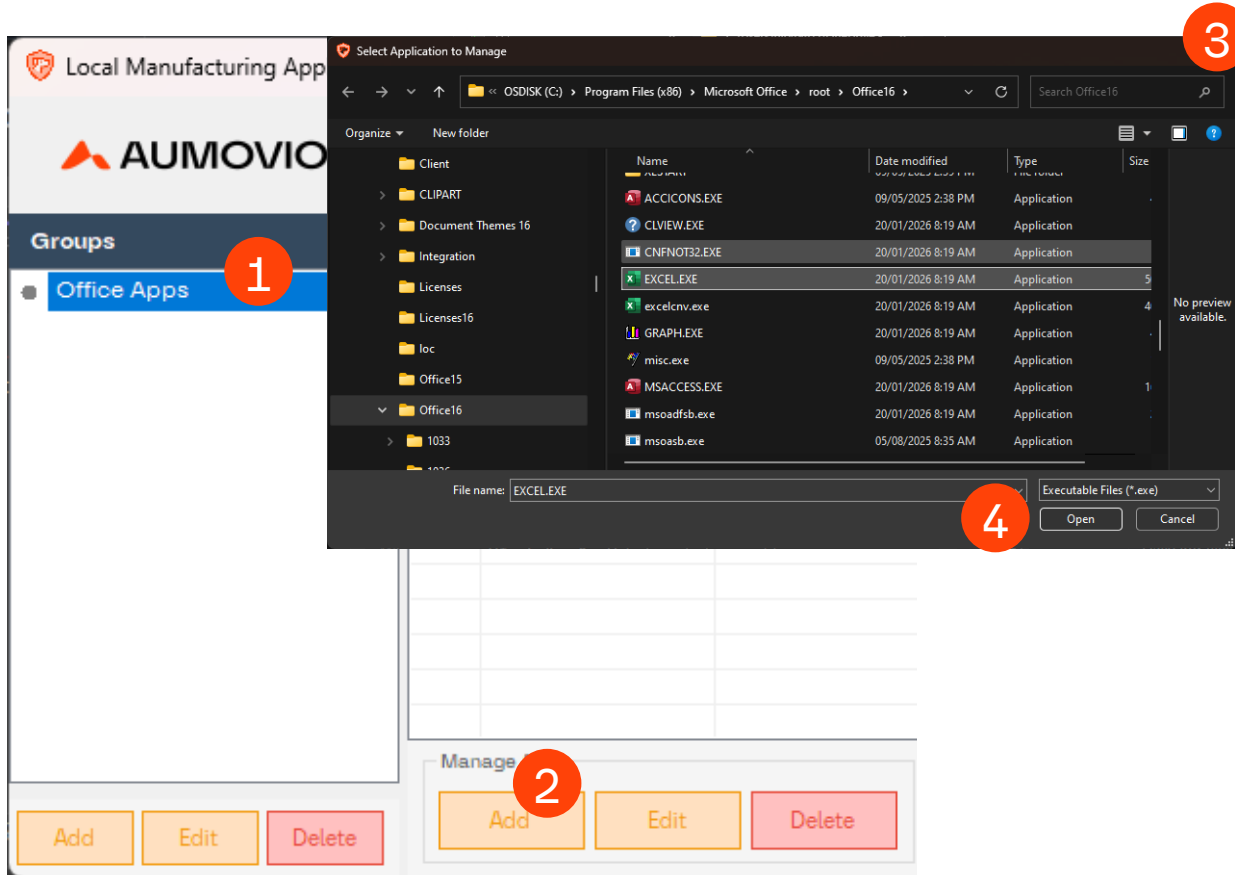
[illegible]

Managing Application for IMEE

Complete Application User Guide

Managing Applications

Let's you organize your applications logically



Adding an Application

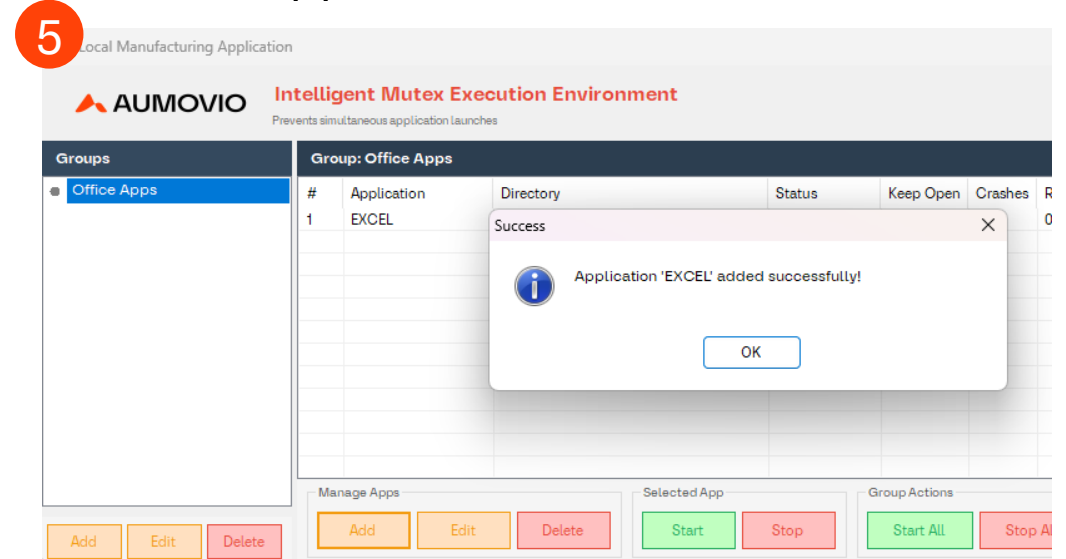
Step 1: Select a group from the left panel

Step 2: Click Add

Step 3: Browse the executable (.exe) file

Step 4: Click open

Step 5: The application appears in the list with status "Stopped"



Let's you organize your applications logically

Edit - WINWORD

Application Path

Microsoft.Office\root\Office16\WINWORD.EXE

Browse

Open

Startup Settings

Startup Delay (sec):

0

Wait time before launching in Start All

Crash Recovery Settings

Keep Open (Auto-Restart): ☐ No

Max Retries:

3

Restart Delay (sec):

5

Stable Run Period (sec):

30

Run time before retry count resets

Health Monitoring

Not Responding (sec):

0

0 = default (2 poll cycles)

Memory Limit (MB):

0

0 = no limit (disabled)

Detect Title Change: ☐ No

KILL if window title changes

Statistics

Crash Count:

0

Reset

Retry Count:

0 / 3

Reset

Last Exit Code:

N/A

4

OK

Cancel

Step 1: Select an application in the list

Step 3: In the edit dialog, you can. See slide 17 to 21 on what each settings mean.

Step 4: Click OK to save changes



Managing Applications

Let's you organize your applications logically

Edit - WINWORD

Application Path

Microsoft Office\root\Office16\WINWORD.EXE

Browse

Open

Startup Settings

Startup Delay (sec): 0

Wait time before launching in Start All

Crash Recovery Settings

Keep Open (Auto-Restart): ☐ No

Max Retries: 3

Restart Delay (sec): 5

Stable Run Period (sec): 30

Run time before retry count resets

Health Monitoring

Not Responding (sec): 0

0 = default (2 poll cycles)

Memory Limit (MB): 0

0 = no limit (disabled)

Detect Title Change: ☐ No

Kill if window title changes

Statistics

Crash Count: 0

Reset

Retry Count: 0 / 3

Reset

Last Exit Code: N/A

OK

Cancel

Application Path

- Directory:** The full path to the executable. Use **Browse** to change it, or **Open** to reveal the file in File Explorer.

Startup Settings

Setting	Range	Default	Description
Startup Delay (sec)	0–300	0	Wait time before launching this app during sequential Start All. Apps with 0 launch immediately.

Crash Recovery Settings

Setting	Range	Default	Description
Keep Open (Auto-Restart)	Yes/No	No	When enabled, the watchdog automatically restarts the app after a crash.

Managing Applications

Let's you organize your applications logically

Edit - WINWORD

Application Path

Microsoft Office\root\Office16\WINWORD.EXE

Browse

Open

Startup Settings

Startup Delay (sec): 0

Wait time before launching in Start All

Crash Recovery Settings

Keep Open (Auto-Restart): ☐ No

Max Retries: 3

Restart Delay (sec): 5

Stable Run Period (sec): 30

Run time before retry count resets

Health Monitoring

Not Responding (sec): 0

0 = default (2 poll cycles)

Memory Limit (MB): 0

0 = no limit (disabled)

Detect Title Change: ☐ No

Kill if window title changes

Statistics

Crash Count: 0

Reset

Retry Count: 0 / 3

Reset

Last Exit Code: N/A

OK

Cancel

Crash Recovery Settings

Setting	Range	Default	Description
Max Retries	1-100	3	Maximum number of consecutive restart attempts before giving up.
Restart Delay (sec)	1-300	5	Seconds to wait before attempting a restart after a crash.
Stable Run Period (sec)	5-600	30	How long the app must run continuously after restart before the retry count resets to 0.

Managing Applications

Let's you organize your applications logically

Edit - WINWORD

Application Path

Microsoft Office\root\Office16\WINWORD.EXE

Browse

Open

Startup Settings

Startup Delay (sec): 0

Wait time before launching in Start All

Crash Recovery Settings

Keep Open (Auto-Restart): ☐ No

Max Retries: 3

Restart Delay (sec): 5

Stable Run Period (sec): 30

Run time before retry count resets

Health Monitoring

Not Responding (sec): 0

0 = default (2 poll cycles)

Memory Limit (MB): 0

0 = no limit (disabled)

Detect Title Change: ☐ No

Kill if window title changes

Statistics

Crash Count: 0

Reset

Retry Count: 0 / 3

Reset

Last Exit Code: N/A

OK

Cancel

Health Monitoring

Setting	Range	Default	Description
Not Responding (sec)	0–600	0	How long a process can remain unresponsive before being force-killed. 0 = default behavior (2 consecutive poll cycles). Set a higher value for apps that legitimately freeze briefly during heavy operations.
Memory Limit (MB)	0–65536	0	Maximum working set memory allowed. If the process exceeds this limit, it is force-killed for auto-restart. 0 = no limit (disabled). Useful for detecting memory leaks in long-running apps.

Managing Applications

Let's you organize your applications logically

Edit - WINWORD

Application Path

Microsoft Office\root\Office16\WINWORD.EXE

Browse

Open

Startup Settings

Startup Delay (sec): 0

Wait time before launching in Start All

Crash Recovery Settings

Keep Open (Auto-Restart): ☐ No

Max Retries: 3

Restart Delay (sec): 5

Stable Run Period (sec): 30

Run time before retry count resets

Health Monitoring

Not Responding (sec): 0

0 = default (2 poll cycles)

Memory Limit (MB): 0

0 = no limit (disabled)

Detect Title Change: ☐ No

Kill if window title changes

Statistics

Crash Count: 0

Reset

Retry Count: 0 / 3

Reset

Last Exit Code: N/A

OK

Cancel

Health Monitoring

Setting	Range	Default	Description
Detect Title Change	Yes/No	No	When enabled, the watchdog records the app's initial window title and treats a persistent title change as a suspected error dialog (e.g., a WinForms unhandled exception dialog whose title is just the app name). Requires 2 consecutive detections to confirm. Leave disabled for apps that legitimately change their window title (e.g., showing a document name).

Managing Applications

Let's you organize your applications logically

Edit - WINWORD

Application Path

Microsoft Office\root\Office16\WINWORD.EXE

Browse

Open

Startup Settings

Startup Delay (sec): 0

Wait time before launching in Start All

Crash Recovery Settings

Keep Open (Auto-Restart): ☐ No

Max Retries: 3

Restart Delay (sec): 5

Stable Run Period (sec): 30

Run time before retry count resets

Health Monitoring

Not Responding (sec): 0

0 = default (2 poll cycles)

Memory Limit (MB): 0

0 = no limit (disabled)

Detect Title Change: ☐ No

Kill if window title changes

Statistics

Crash Count: 0

Reset

Retry Count: 0 / 3

Reset

Last Exit Code: N/A

OK

Cancel

Statistics	
Field	Description
Crash Count	Total number of detected crashes (cumulative). Click Reset to clear
Retry Count	Current consecutive restart attempt count vs. max retries (e.g., "1 / 3"). Click Reset to clear and remove "Failed" state.
Last Exit Code	The exit code from the last process termination. N/A if no exit has been recorded. A non-zero exit code (shown in red) typically indicates the process crashed or terminated with an error.

Managing Applications

Let's you organize your applications logically

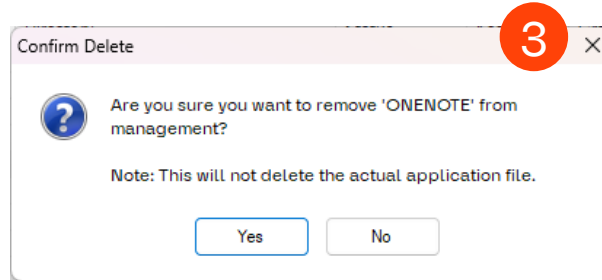
Group: Office365 Apps		
#	Application	Directory
1	EXCEL	C:\Program Files (x86
2	MSACCESS	C:\Program Files (x86
3	WINWORD	C:\Program Files (x86
4	MSPUB	C:\Program Files (x86
5	ONENOTE	C:\Program Files (x86

Manage Apps

Add

Edit

Delete



Deleting an Application

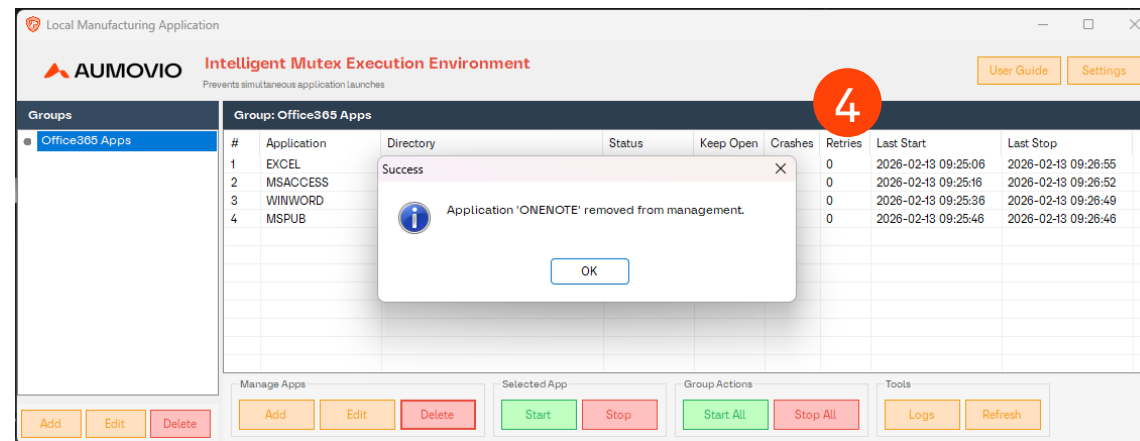
Step 1: Select an application in the list

Step 2: Click Delete

Step 3: Confirm deletion

Step 4: The application is removed from the management

Note: The actual executable file is not deleted from disk



Starting and Stopping Application

Complete Application User Guide

Starting and Stopping Application

Starting a Single Application

Group: Office365 Apps			
#	Application	Directory	Status
1	EXCEL	C:\Program Files (x86)\Microsoft Offic...	Stopped
2	MSACCESS	C:\Program Files (x86)\Microsoft Offic...	Stopped
3	WINWORD	C:\Program Files (x86)\Microsoft Offic...	Stopped
4	MSPUB	C:\Program Files (x86)\Microsoft Offic...	Stopped

Manage Apps

AddEditDelete

Selected

StartStop

Group: Office365 Apps			
#	Application	Directory	Status
1	EXCEL	C:\Program Files (x86)\Microsoft Offic...	Starting...
2	MSACCESS	C:\Program Files (x86)\Microsoft Offic...	Stopped
3	WINWORD	C:\Program Files (x86)\Microsoft Offic...	Stopped
4	MSPUB	C:\Program Files (x86)\Microsoft Offic...	Stopped

Manage Apps

AddEditDelete

Selected App

StartStop

Start an Application

Step 1: Select an application from the list

Step 2: Click Start

Step 3: The status changes to **Starting...** (orange)

Step 4: Once the application window appears, the status changes to Running (green)

Step 5: The “Last Start” timestamp is recorded

Local Manufacturing Application									
AUMOVIO Intelligent Mutex Execution Environment									
Prevents simultaneous application launches									
Groups									
Office365 Apps 1 / 4									
#	Application	Directory	Status	Keep Open	Crashes	Retries	Last Start	Last Stop	
1	EXCEL	C:\Program Files (x86)\Microsoft Offic...	Running	Yes	1	0	2026-02-13 13:53:47	2026-02-13 09:26:55	
2	MSACCESS	C:\Program Files (x86)\Microsoft Offic...	Stopped	Yes	0	0	2026-02-13 09:25:16	2026-02-13 09:26:52	
3	WINWORD	C:\Program Files (x86)\Microsoft Offic...	Stopped	Yes	0	0	2026-02-13 09:25:36	2026-02-13 09:26:49	
4	MSPUB	C:\Program Files (x86)\Microsoft Offic...	Stopped	Yes	0	0	2026-02-13 09:25:46	2026-02-13 09:26:46	

Manage Apps

AddEditDelete

Selected App

StartStop

Group Actions

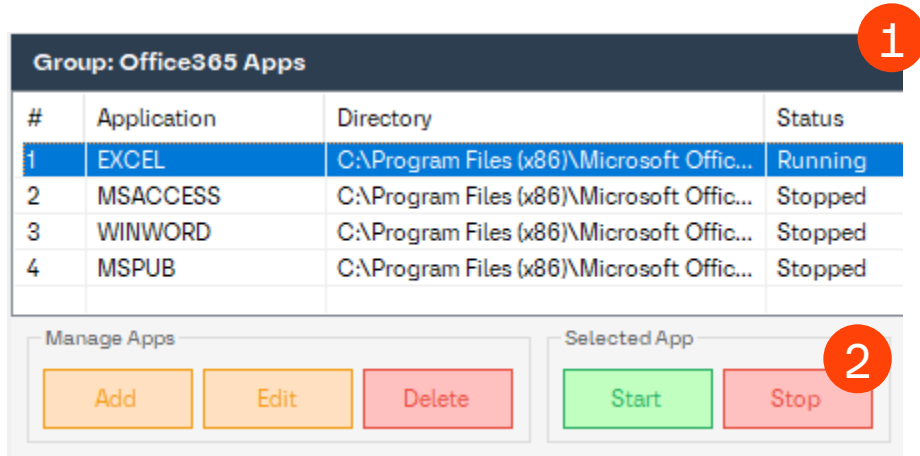
Start AllStop All

Tools

LogsRefresh

Starting and Stopping Application

Stopping a Single Application



Stop an Application

Step 1: Select a running application

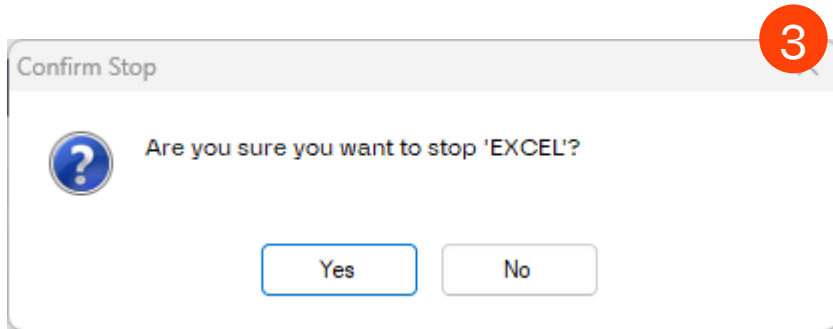
Step 2: Click Stop

Step 3: Confirm the action

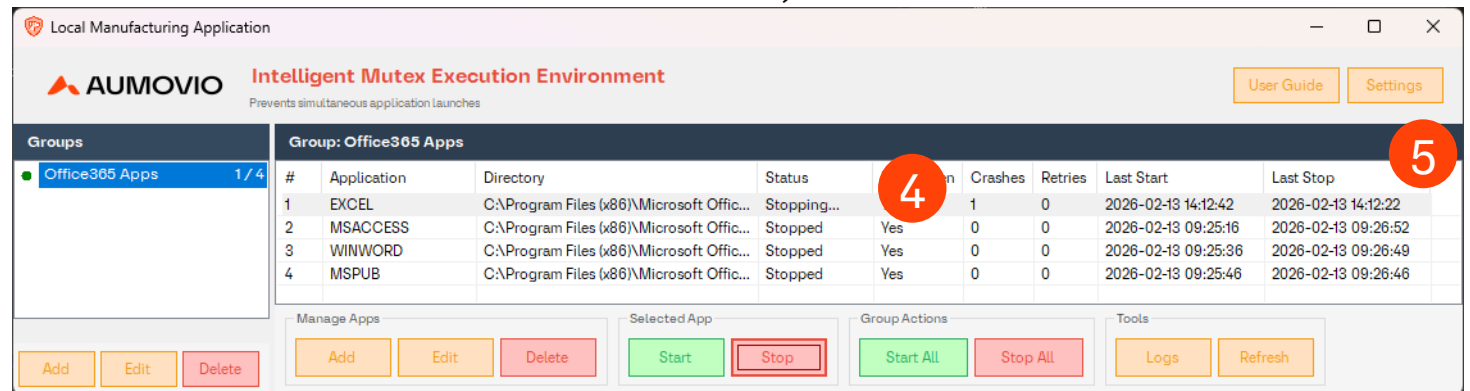
Step 4: The status changes to **Stopping...** (orange)

Step 5: Once the process exits, the status changes to **Stopped** (black)

Step 6: The "Last Stop" timestamp is recorded

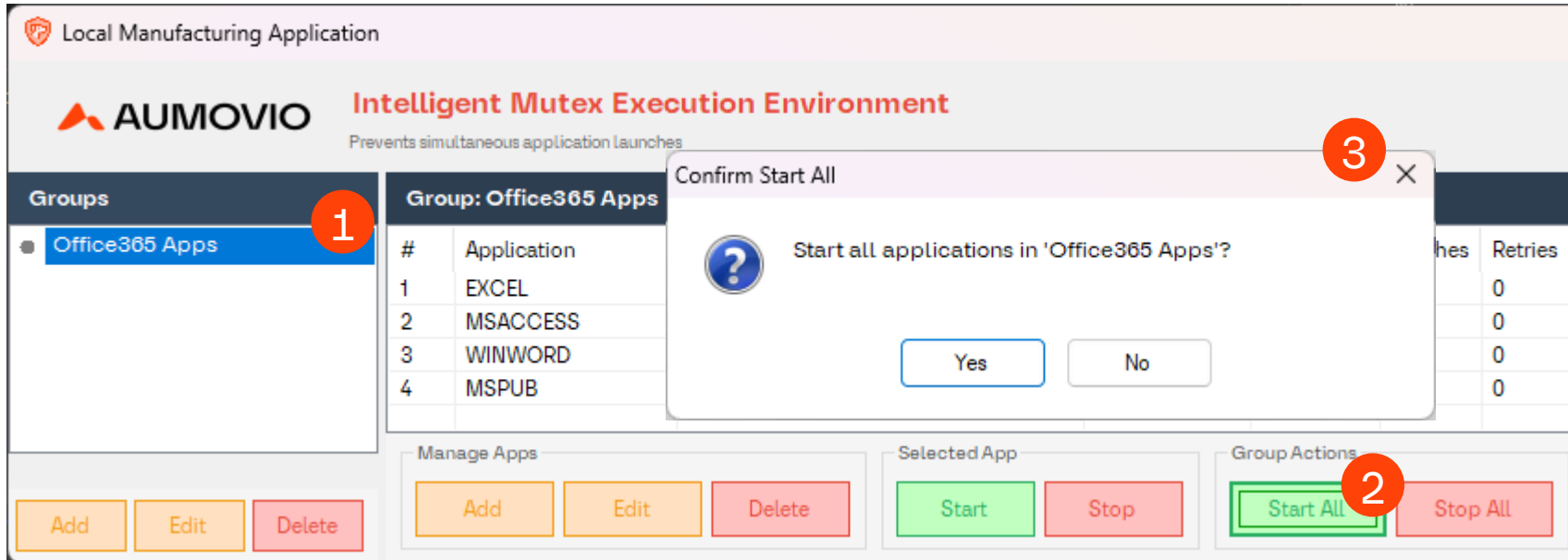


IMEE attempts a graceful shutdown first. If the application doesn't close within 3 seconds, it is force-killed.



Starting and Stopping Application

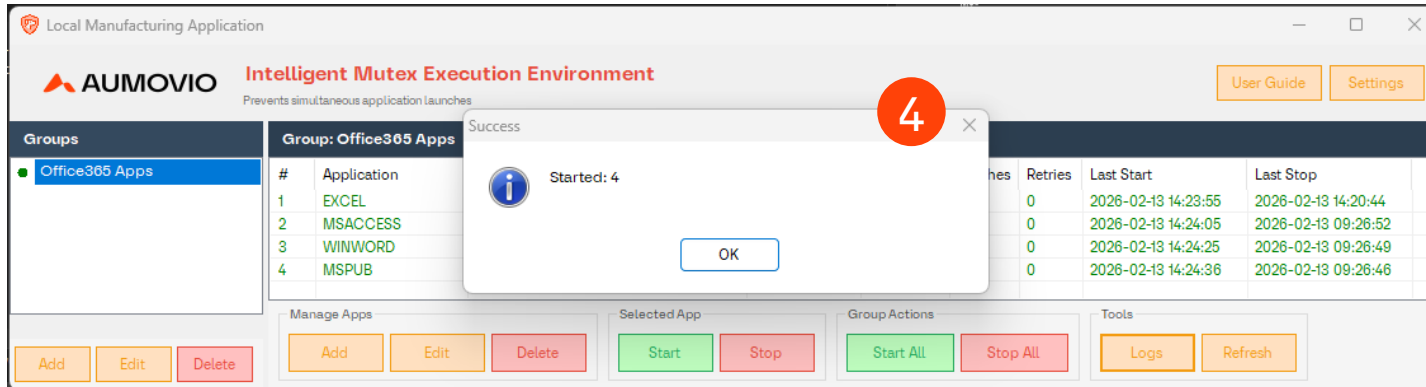
Starting All Application



Starting All Application

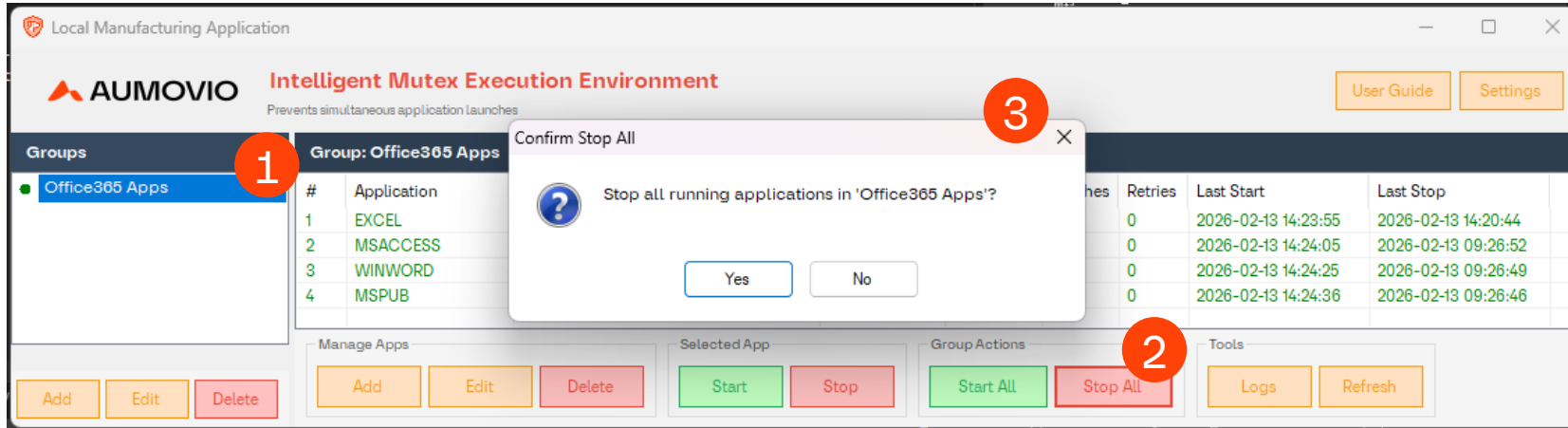
Step 1: Select a Group
Step 2: Click Start All
Step 3: Confirm the action
Step 4: All stopped applications in the group are launched

With Startup Delays: If any applications have a Startup Delay configured (via Edit), they launch one at a time. Each app shows a "Waiting (Ns)" countdown before it launches. This prevents overloading the system when starting many apps at once.



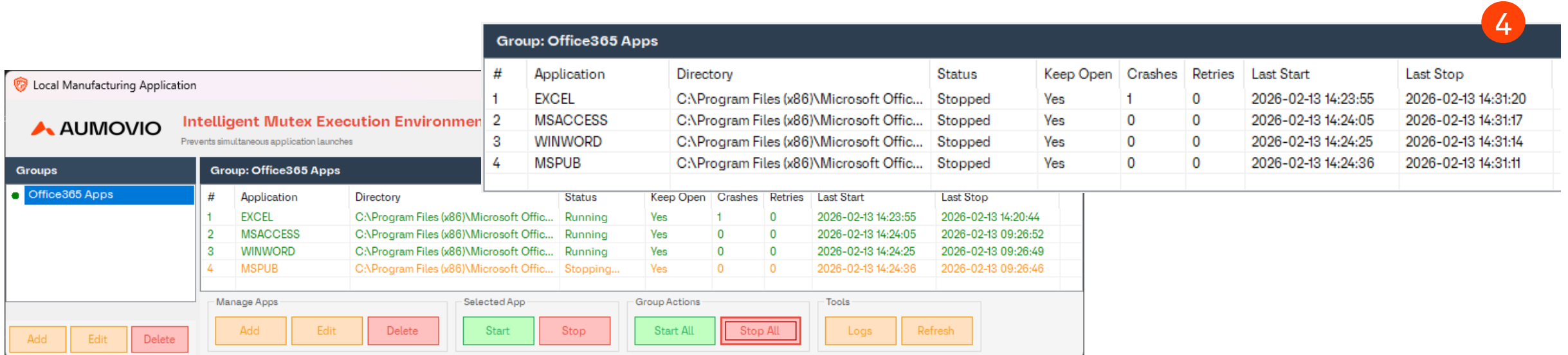
Starting and Stopping Application

Stopping All Application



Stopping All Application

Step 1: Select a Group
Step 2: Click Stop All
Step 3: Confirm the action
Step 4: All running applications in the group are stopped



Keep Open (Auto-Restart)

Complete Application User Guide

Keep Open (Auto-Restart)

Keep Open automatically restarts an application when it crashes or is closed unexpectedly.

Group: Office365 Apps		
#	Application	Directory
1	EXCEL	C:\Program Files (x86
2	MSACCESS	C:\Program Files (x86
3	WINWORD	C:\Program Files (x86
4	MSPUB	C:\Program Files (x86
5	ONENOTE	C:\Program Files (x86

Manage Apps

AddEditDelete

Edit - EXCEL

Application Path

36)\Microsoft Office\root\Office16\EXCELEXE

BrowseOpen

Startup Settings

Startup Delay (sec): 0

Wait time before launching in Start All

Crash Recovery Settings

Keep Open (Auto-Restart): ☒ Yes

Max Retries: 3

Restart Delay (sec): 5

Stable Run Period (sec): 30

Run time before retry count resets

Health Monitoring

Not Responding (sec): 0

0 = default (2 poll cycles)

Memory Limit (MB): 0

0 = no limit (disabled)

Detect Title Change: ☐ No

Kill if window title changes

Statistics

Crash Count: 0

Reset

Retry Count: 0 / 3

Reset

Last Exit Code: N/A

OKCancel

Enabling Keep Open

Step 1: Select an application in the list

Step 2: Click Edit

Step 3: Configure the settings:

- Check Keep Open
- Restart Delay: How long to wait before restarting (default: 5 seconds)
- Max Retries: How many times to attempt restarting (default: 3)
- Stable Run Period: How long the app must run before retry count resets

Step 4: Click OK

Local Manufacturing Application

AUMOVIO Intelligent Mutex Execution Environment

Prevents simultaneous application launches

User GuideSettings

Success

Application updated successfully!

OK

Groups	#	Application	Retries	Last Start	Last Stop
Office365 Apps	1	EXCEL	0	2026-02-13 14:23:55	2026-02-13 14:31:20
	2	MSACCESS	0	2026-02-13 14:24:05	2026-02-13 14:31:17
	3	WINWORD	0	2026-02-13 14:24:25	2026-02-13 14:31:14
	4	MSPUB	0	2026-02-13 14:24:36	2026-02-13 14:31:11

Manage Apps

AddEditDelete

Selected App

Group Actions

StartStopStart AllStop All

Tools

LogsRefresh

Keep Open (Auto-Restart)

How it works

Local Manufacturing Application

 **AUMOVIO**

Intelligent Mutex Execution Environment

Prevents simultaneous application launches

User Guide

Settings

Groups

Office365 Apps

Group: Office365 Apps									
#	Application	Directory	Status	Keep Open	Crashes	Retries	Last Start	Last Stop	
1	EXCEL	C:\Program Files (x86)\Microsoft Office...	Restarting (5s)	Yes	6	2	2026-02-17 08:39:03	2026-02-17 08:39:23	
2	MSACCESS	C:\Program Files (x86)\Microsoft Office...	Running	Yes	3	0	2026-02-17 08:36:10	2026-02-13 15:30:05	
3	WINWORD	C:\Program Files (x86)\Microsoft Office...	Running	Yes	0	0	2026-02-17 08:36:30	2026-02-13 15:30:02	
4	MSPUB	C:\Program Files (x86)\Microsoft Office...	Running	Yes	0	0	2026-02-17 08:36:41	2026-02-13 15:29:59	
									</

Local Manufacturing Application

AUMOVIO

Intelligent Mutex Execution Environment

Prevents simultaneous application launches

User Guide

Settings

Groups

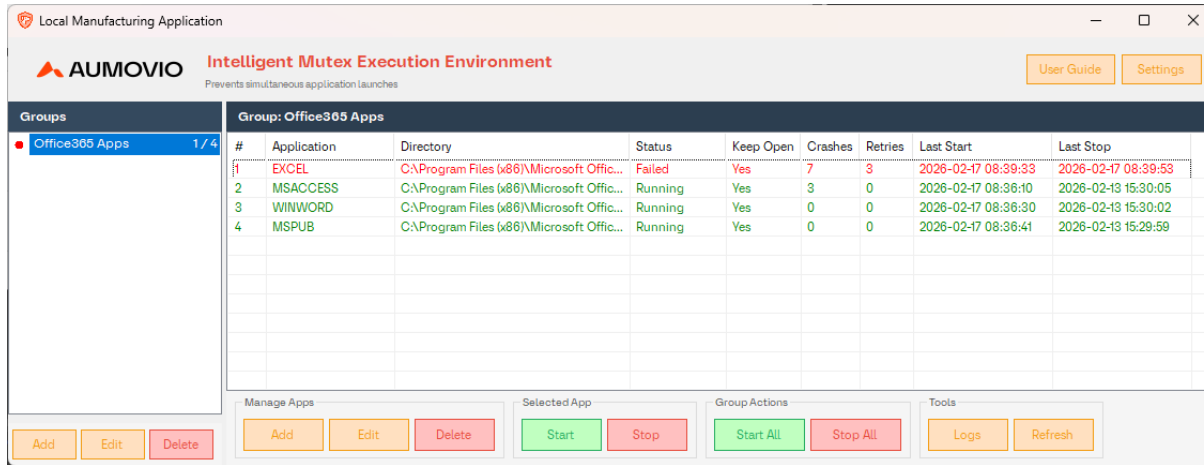
Office365 Apps

Group: Office365 Apps

#	Application	Directory	Status	Keep Open	Crashes	Retries	Last Start	Last Stop
1	EXCEL	C:\Program Files (x86)\Microsoft Office\...	Starting...	Yes	6	2	2026-02-17 08:39:33	2026-02-17 08:39:23
2	MSACCESS	C:\Program Files (x86)\Microsoft Office\...	Running	Yes	3	0	2026-02-17 08:36:10	2026-02-13 15:30:05
3	WINWORD	C:\Program Files (x86)\Microsoft Office\...	Running	Yes	0	0	2026-02-17 08:36:30	2026-02-13 15:30:02
4	MSPUB	C:\Program Files (x86)\Microsoft Office\...	Running	Yes	0	0	2026-02-17 08:36:41	2026-02-13 15:29:59

- When a Keep Open application stops unexpectedly:**
1. The crash is detected by the watchdog (within one poll interval, default: 10 seconds)
 2. The crash count increments
 3. The retry count increments
 4. A countdown begins: **Restarting (5s) ? Restarting (4s) ? ... ? Starting...**
 5. The application relaunches automatically
 6. If the relaunch succeeds and the app runs stably for the **Stable Run Period** (default: 30 seconds), the retry count resets to 0

Keep Open (Auto-Restart) When Max Retries Are Exhausted



Local Manufacturing Application

AUMOVIO Intelligent Mutex Execution Environment

Prevents simultaneous application launches

User Guide Settings

Groups

Office365 Apps 1/4

#	Application	Directory	Status	Keep Open	Crashes	Retries	Last Start	Last Stop
1	EXCEL	C:\Program Files (x86)\Microsoft Office...	Failed	Yes	7	3	2026-02-17 08:39:33	2026-02-17 08:39:53
2	MSACCESS	C:\Program Files (x86)\Microsoft Office...	Running	Yes	3	0	2026-02-17 08:36:10	2026-02-13 15:30:05
3	WINWORD	C:\Program Files (x86)\Microsoft Office...	Running	Yes	0	0	2026-02-17 08:36:30	2026-02-13 15:30:02
4	MSPUB	C:\Program Files (x86)\Microsoft Office...	Running	Yes	0	0	2026-02-17 08:36:41	2026-02-13 15:29:59

Manage Apps

Add Edit Delete

Selected App

Start Stop

Group Actions

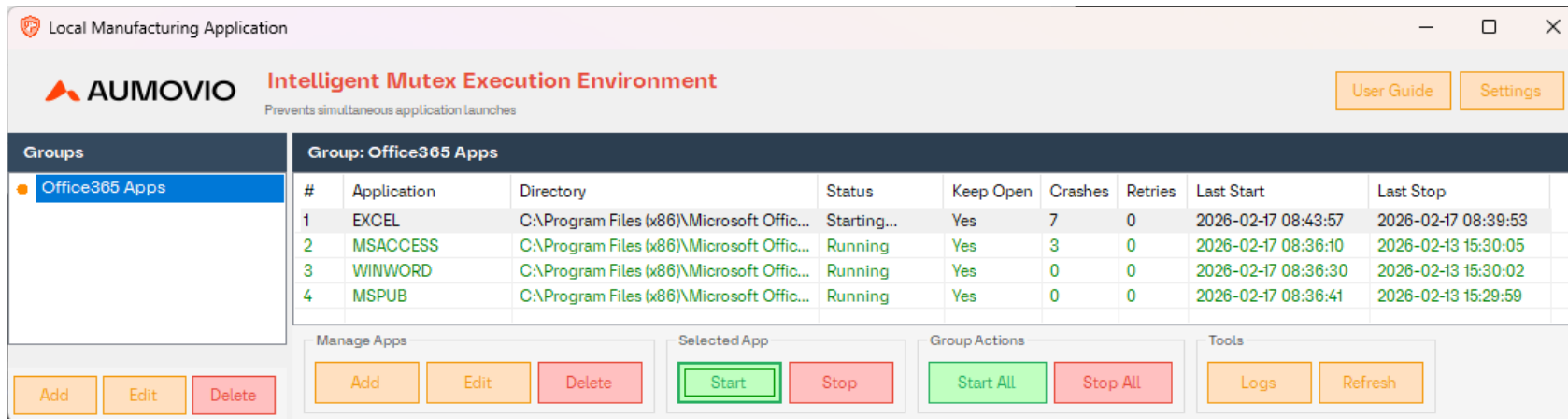
Start All Stop All

Tools

Logs Refresh

If the application crashes more times than the max retry limit (without running stably between crashes):

1. The status changes to **Failed** (red)
2. No further auto-restart attempts are made
3. You can fix the issue and either:
 - Click **Start** to manually restart (resets the retry count)
 - Click **Edit** and reset the retry count, then start



Local Manufacturing Application

AUMOVIO Intelligent Mutex Execution Environment

Prevents simultaneous application launches

User Guide Settings

Groups

Office365 Apps

#	Application	Directory	Status	Keep Open	Crashes	Retries	Last Start	Last Stop
1	EXCEL	C:\Program Files (x86)\Microsoft Office...	Starting...	Yes	7	0	2026-02-17 08:43:57	2026-02-17 08:39:53
2	MSACCESS	C:\Program Files (x86)\Microsoft Office...	Running	Yes	3	0	2026-02-17 08:36:10	2026-02-13 15:30:05
3	WINWORD	C:\Program Files (x86)\Microsoft Office...	Running	Yes	0	0	2026-02-17 08:36:30	2026-02-13 15:30:02
4	MSPUB	C:\Program Files (x86)\Microsoft Office...	Running	Yes	0	0	2026-02-17 08:36:41	2026-02-13 15:29:59

Manage Apps

Add Edit Delete

Selected App

Start Stop

Group Actions

Start All Stop All

Tools

Logs Refresh

Keep Open (Auto-Restart)

Stable Run Period

The retry count only resets to 0 after the application has been running continuously for the configured **Stable Run Period** (default: 30 seconds). This prevents apps that crash shortly after starting from resetting retries indefinitely and never reaching "Failed" state.

You can configure this per-application via **Edit ? Stable Run Period (seconds)**. [see slide 29]

The screenshot shows the AUMOVIO Intelligent Mutex Execution Environment interface. The title bar reads "Local Manufacturing Application". The header includes the AUMOVIO logo, the text "Intelligent Mutex Execution Environment", and the subtitle "Prevents simultaneous application launches". There are "User Guide" and "Settings" buttons in the top right.

On the left, a "Groups" sidebar shows "Office365 Apps" selected. The main area displays a table for the "Group: Office365 Apps".

#	Application	Directory	Status	Keep Open	Crashes	Retries	Last Start	Last Stop
1	EXCEL	C:\Program Files (x86)\Microsoft Offic...	Starting...	Yes	7	0	2026-02-17 08:43:57	2026-02-17 08:39:53
2	MSACCESS	C:\Program Files (x86)\Microsoft Offic...	Running	Yes	3	0	2026-02-17 08:36:10	2026-02-13 15:30:05
3	WINWORD	C:\Program Files (x86)\Microsoft Offic...	Running	Yes	0	0	2026-02-17 08:36:30	2026-02-13 15:30:02
4	MSPUB	C:\Program Files (x86)\Microsoft Offic...	Running	Yes	0	0	2026-02-17 08:36:41	2026-02-13 15:29:59

At the bottom, there are several control panels:

- Manage Apps:** Add, Edit, Delete buttons.
- Selected App:** Start (green), Stop (red) buttons.
- Group Actions:** Start All (green), Stop All (red) buttons.
- Tools:** Logs, Refresh buttons.

Keep Open (Auto-Restart)

Unresponsive Application Detection

If a Keep Open application becomes unresponsive (e.g., due to an unhandled exception dialog that blocks the main window), Instance Manager will detect it and force-kill the process so auto-restart can take over. The detection requires two consecutive poll cycles to confirm the app is truly unresponsive, avoiding false positives from brief UI freezes.

Important Notes

- Manually stopping an app via the **Stop** button does **not** trigger a restart – even with Keep Open enabled
- The retry count resets to 0 whenever the app successfully starts (manual or auto)
- The crash count is cumulative and never resets automatically (reset it via Edit)

The screenshot displays the AUMOVIO Intelligent Mutex Execution Environment interface. The title bar reads "Local Manufacturing Application". The header includes the AUMOVIO logo, the text "Intelligent Mutex Execution Environment", and a subtitle "Prevents simultaneous application launches". There are "User Guide" and "Settings" buttons in the top right.

The main content area features a table titled "Group: Office365 Apps". The table has columns for #, Application, Directory, Status, Keep Open, Crashes, Retries, Last Start, and Last Stop. The data rows are as follows:

#	Application	Directory	Status	Keep Open	Crashes	Retries	Last Start	Last Stop
1	EXCEL	C:\Program Files (x86)\Microsoft Offic...	Starting...	Yes	7	0	2026-02-17 08:43:57	2026-02-17 08:39:53
2	MSACCESS	C:\Program Files (x86)\Microsoft Offic...	Running	Yes	3	0	2026-02-17 08:36:10	2026-02-13 15:30:05
3	WINWORD	C:\Program Files (x86)\Microsoft Offic...	Running	Yes	0	0	2026-02-17 08:36:30	2026-02-13 15:30:02
4	MSPUB	C:\Program Files (x86)\Microsoft Offic...	Running	Yes	0	0	2026-02-17 08:36:41	2026-02-13 15:29:59

Below the table, there are several control sections: "Manage Apps" with "Add", "Edit", and "Delete" buttons; "Selected App" with "Start" and "Stop" buttons; "Group Actions" with "Start All" and "Stop All" buttons; and "Tools" with "Logs" and "Refresh" buttons.

Health Monitoring

Complete Application User Guide

Health Monitoring

Not Responding Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Not Responding Detection

Detects when an application's UI thread is frozen (e.g., infinite loop, UI deadlock, blocked message pump).

How it works:

1. Each poll tick checks `Process.Responding` for all windowed processes
2. If an app is not responding, tracking begins
3. After the configured timeout (or 2 consecutive poll cycles if timeout is 0), the process is force-killed
4. Auto-restart takes over

Configure via: Edit → **Not Responding (sec)**

Tip: Set a non-zero value (e.g., 30 seconds) for apps that occasionally freeze briefly during heavy operations like loading large files.

Health Monitoring

Error Dialog Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Error Dialog Detection

Detects when an application shows a crash or error dialog that is technically "responding" (pumping messages) but has the app effectively stuck. This catches error dialogs that `Process.Responding` cannot detect because the dialog's message pump keeps the window alive.

Detected patterns include:

- "has stopped working", "has encountered a problem"
- "unhandled exception", "application error", "runtime error", "fatal error"
- ".NET Framework", "CLR error", "just-in-time debugging"
- "assertion failed", "access violation", "stack overflow"
- .NET exception type names

(e.g., `NullReferenceException`, `DivideByZeroException`, `OutOfMemoryException`, `StackOverflowException`, `AccessViolationException`, `InvalidOperationException`)

- Generic phrases like "an error occurred", "error in application"

Health Monitoring

Error Dialog Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

How it works:

1. Each poll tick checks the main window title against known error dialog patterns
2. If a match is found, it waits one more poll cycle to confirm (avoids false positives)
3. On second consecutive detection, the process is force-killed
4. Auto-restart takes over

This detection is **always active** for Keep Open apps – no configuration needed.

Health Monitoring

Window Title Change Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Window Title Change Detection

Detects when an application's window title changes unexpectedly, which can indicate a WinForms ThreadExceptionDialog or similar error dialog whose title doesn't contain any recognizable error keywords.

How it works:

1. When an app first appears as running, its window title is recorded as the baseline
2. Each poll tick compares the current title to the baseline
3. If the title changes and stays changed for 2 consecutive poll cycles, the process is force-killed
4. Auto-restart takes over

Configure via: Edit → **Detect Title Change** (opt-in, disabled by default)

Warning: Only enable this for apps with a stable, unchanging window title. Apps that show document names, status text, or progress in their title bar will trigger false positives.

Health Monitoring

Memory Limit Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Memory Limit Detection

Detects when an application's working set memory exceeds a configured threshold. Useful for catching memory leaks in long-running 24/7 applications.

How it works:

1. Each poll tick reads `Process.WorkingSet64` (a lightweight kernel query with no overhead)
2. If the working set exceeds the configured limit, the process is force-killed immediately (no confirmation wait)
3. Auto-restart takes over

Configure via: Edit → **Memory Limit (MB)** (set to 0 to disable)

Tip: Monitor your app's normal memory usage first, then set the limit to 2–3× the expected peak. For example, if an app normally uses 200 MB, set the limit to 500 MB.

Health Monitoring

CPU-Hung Process Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

CPU-Hung Process Detection

Detects when an application is consuming excessive CPU continuously, indicating an infinite loop or spin-wait deadlock.

How it works:

1. Each poll tick samples `Process.TotalProcessorTime` (a lightweight kernel query)
2. CPU utilization is computed by comparing samples across ticks: $\Delta \text{CpuTime} / (\Delta \text{time} \times \text{ProcessorCount})$
3. If CPU usage exceeds 95% for 3 consecutive poll cycles, the process is force-killed
4. Brief CPU spikes are tolerated – only sustained high CPU triggers the kill

This detection is **always active** for Keep Open apps – no configuration needed.

Note: This catches spin-wait deadlocks (high CPU) but not idle deadlocks (0% CPU with frozen UI). Idle deadlocks are caught by Not Responding Detection instead.

Health Monitoring

Background Zombie Detection

Instance Manager includes several health monitoring mechanisms that detect unhealthy applications beyond simple crash detection. All health monitoring features apply only to **Keep Open** applications and trigger the same auto-restart flow as a regular crash.

Background Zombie Detection

Detects when an application's window disappears but its process remains alive in the background (e.g., Excel closing its window but lingering as a background process).

How it works:

1. If the watchdog detects that an app was running (had a window) but now has no window, it checks for background processes with the same name
2. Any found background processes are force-killed
3. The app is then treated as crashed and the normal restart flow begins

This detection is **always active** – no configuration needed.

Health Monitoring

Detection Coverage Summary

The following table shows which types of application failures Instance Manager can detect and handle

Failure Type	Example	Detection Method	Auto-Restart?
UI Freeze	Infinite loop on UI thread	Not Responding detection	Yes
Unhandled Exception Dialog	NullPointerException, DivideByZeroException on UI thread	Error Dialog title matching + Title Change detection	Yes
Stack Overflow	Infinite recursion	Process terminates → crash detection	Yes
Out of Memory	Memory exhaustion	Memory Limit detection (proactive) + crash detection (reactive)	Yes
Access Violation	Writing to invalid memory	Process terminates → crash detection	Yes

Health Monitoring

Detection Coverage Summary

The following table shows which types of application failures Instance Manager can detect and handle

Failure Type	Example	Detection Method	Auto-Restart?
UI Deadlock	Two locks acquired in opposite order	Not Responding detection (UI blocked) + CPU-Hung detection (spin-wait)	Yes
Background Thread Exception	Unhandled exception on worker thread	Process terminates → crash detection + zombie detection	Yes
Environment.FailFast	Immediate process termination	Process terminates → crash detection	Yes
Process Crash	Any unexpected process exit	Window disappears → crash detection	Yes
Memory Leak	Gradual memory growth	Memory Limit detection	Yes

Health Monitoring

Detection Coverage Summary

The following table shows which types of application failures Instance Manager can detect and handle

Failure Type	Example	Detection Method	Auto-Restart?
CPU Spin Loop	Infinite while(true) with work	CPU-Hung detection (3 consecutive ticks >95%)	Yes
Zombie Process	Window closed but process lingers	Background Zombie detection	Yes

User Experience

Complete Application User Guide

Understanding Status Indicators

Application Status

Status	Color	What it Means	What to Do
Stopped	Black	Application is not Running	Click Start to launch
Starting...	Orange	Application was just launched, waiting for its window to appear	Wait – will change to Running
Running	Green	Application is running normally	No action needed
Running (Other Group)	Dark Cyan	Application is running, but it was started from a different group	Stop it in the other group first if you want to start it here
Stopping...	Orange	Stop command was sent, waiting for the process to exit	Wait – will change to Stopped
Restarting (Ns)	Orange	Application crashed, auto-restart countdown in progress	Wait for countdown to finish
Starting...	Orange	Restart countdown finished, application is being relaunched	Wait – will change to Running

Understanding Status Indicators

Application Status

Status	Color	What it Means	What to Do
Waiting (Ns)	Orange	Sequential Start All – this app is next in line	Wait for countdown to finish
Failed	Red	Auto-restart gave up after max retries	Check the app, then Start manually or reset retries

Note: Keep Open applications that become unresponsive (e.g., stuck on an unhandled exception dialog) are automatically detected and force-killed after two consecutive poll cycles, allowing auto-restart to take over.

Understanding Status Indicators

Group Status

Each group in the left panel displays a colored circle indicator that summarizes the state of all applications in that group. When only some applications are in a given state, a count suffix (e.g., "2 / 5") appears next to the group name.

Color	What it Means	Count suffix
Gray	Group has no applications, or all applications are stopped	N/A
Green	All applications are running	N/A
Green	Some applications are running	N / T (running / total)
Orange	One or more applications are in a transitional state (Starting, Stopping, Restarting or Waiting)	N/A
Red	All applications have failed (max retries exhausted)	N/A
Red	Some applications have failed	N / T (failed / total)

Priority: If a group has apps in multiple states, the indicator shows the highest-priority state: Red (failed) > Orange (transitional) > Green (running) > Gray (stopped/empty).



Settings

Complete Application User Guide

Settings

IMEE Settings and Performance tweaks

Settings

General

Station Name:

Performance

Status Poll Interval (ms):

10000

1000 - 60000

Start Grace Period (seconds):

10

1 - 120

GC Collect Interval (minutes):

30

5 - 1440

Storage Save Interval (seconds):

30

5 - 300

Logging

Log Flush Interval (seconds):

10

1 - 120

Log Buffer Size (entries):

100

10 - 1000

Log Retention (days):

7

1 - 365

Reset Defaults

OK

Cancel

The Settings dialog lets you configure the station name and tune performance and logging parameters. Click the **Settings** button in the header to open it.

General

Setting	Description	Default
Station Name	Identifies the workstation in the title bar	(empty

Logging

Setting	Range	Default	Description
Log Flush Interval (seconds)	1 – 120	10	How often buffered log entries are written to disk
Log Buffer Size (entries)	10 – 1000	100	Maximum buffered entries before a flush is forced.
Log Retention (days)	1 – 365	7	Log files older than this are automatically deleted

Settings

IMEE Settings and Performance tweaks

Settings

General

Station Name:

Performance

Status Poll Interval (ms):

10000

1000 - 60000

Start Grace Period (seconds):

10

1 - 120

GC Collect Interval (minutes):

30

5 - 1440

Storage Save Interval (seconds):

30

5 - 300

Logging

Log Flush Interval (seconds):

10

1 - 120

Log Buffer Size (entries):

100

10 - 1000

Log Retention (days):

7

1 - 365

Reset Defaults

OK

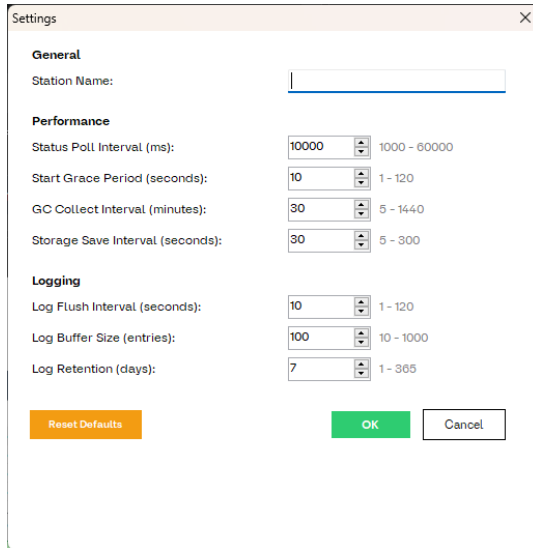
Cancel

Performance

Setting	Range	Default	Description
Status Poll Interval (milliseconds)	1000 – 60000	10000	How often the watchdog checks process statuses. Lower = more responsive but higher CPU usage.
Start Grace Period (seconds)	1 – 120	10	Time after launching an app before the watchdog starts monitoring it. Allows the window to appear.
GC Collect Interval (minutes)	5 – 1440	30	How often a periodic garbage collection runs to prevent long-term memory growth during 24/7 operation.
Storage Save Interval	5 – 300	30	Minimum time between throttled disk writes for timer-tick updates. User-initiated saves (Add, Edit, Delete) always write immediately.

Settings

IMEE Settings and Performance tweaks



The screenshot shows a 'Settings' dialog box with a title bar and a close button. It is divided into three sections: General, Performance, and Logging. The General section has a 'Station Name' text input field. The Performance section has four spinners: 'Status Poll Interval (ms)' with a range of 1000 to 60000, 'Start Grace Period (seconds)' with a range of 1 to 120, 'GC Collect Interval (minutes)' with a range of 5 to 1440, and 'Storage Save Interval (seconds)' with a range of 5 to 300. The Logging section has three spinners: 'Log Flush Interval (seconds)' with a range of 1 to 120, 'Log Buffer Size (entries)' with a range of 10 to 1000, and 'Log Retention (days)' with a range of 1 to 365. At the bottom, there are three buttons: 'Reset Defaults' (orange), 'OK' (green), and 'Cancel' (white).

Section	Setting	Value	Range
General	Station Name		
Performance	Status Poll Interval (ms)	10000	1000 - 60000
	Start Grace Period (seconds)	10	1 - 120
	GC Collect Interval (minutes)	30	5 - 1440
	Storage Save Interval (seconds)	30	5 - 300
Logging	Log Flush Interval (seconds)	10	1 - 120
	Log Buffer Size (entries)	100	10 - 1000
	Log Retention (days)	7	1 - 365

Reset Defaults

Click **Reset Defaults** in the Settings dialog to restore all performance and logging settings to their defaults. The station name is not affected by reset.

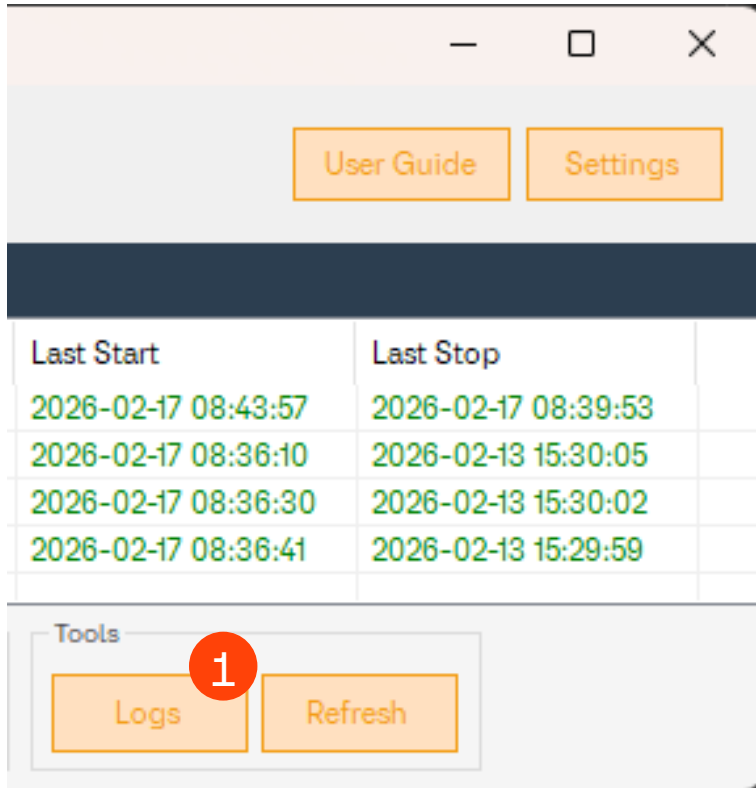
All settings are saved in `settings.json` and persist across restarts. Changes take effect immediately – no restart required.

Logs

Complete Application User Guide

Logs

Instance Manager logs all operations to daily log files.

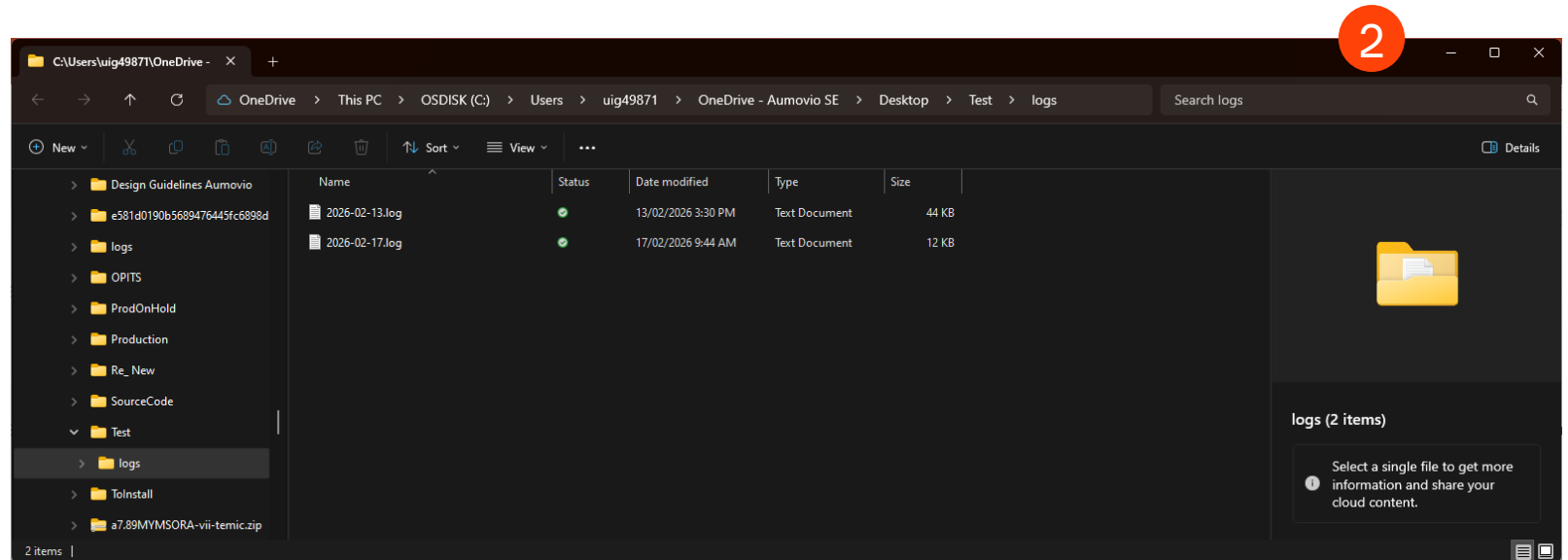


Viewing Logs

Step 1: Click the **Logs** button to open the logs folder in File Explorer

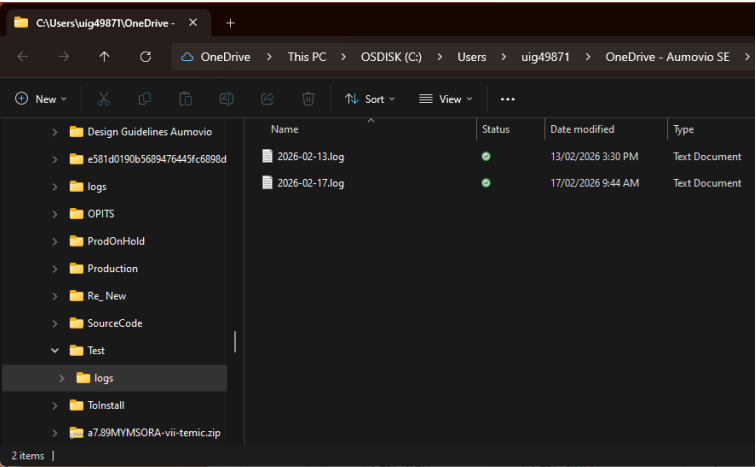
Step 2: You can now browse the logs

Log files are named `YYYY-MM-DD.log`



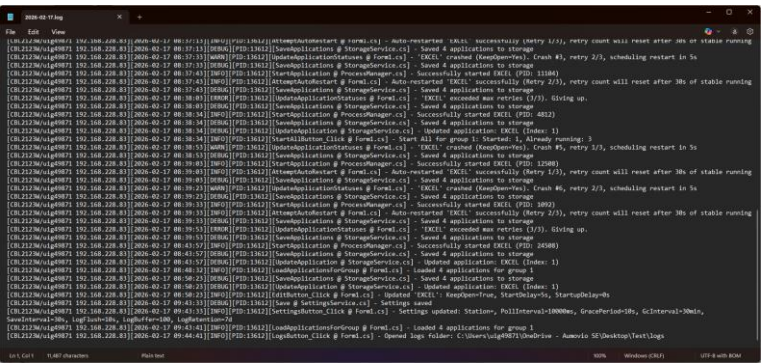
Logs

Instance Manager logs all operations to daily log files.



What Gets Logged

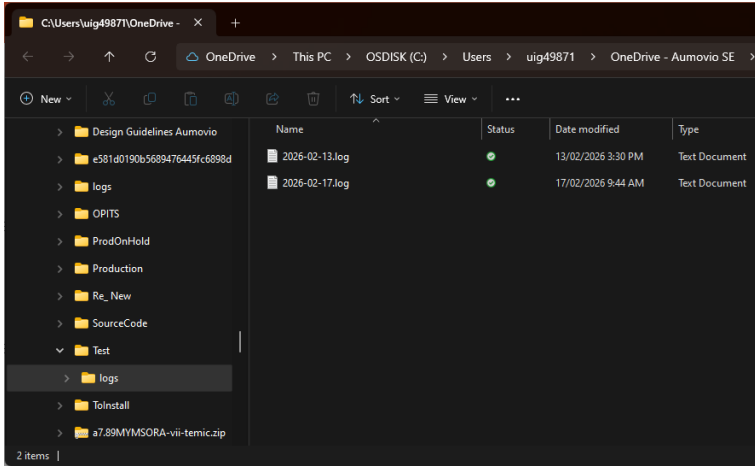
- Application starts and stops (including PIDs)
- Unauthorized launch detections
- Crash detections and auto-restart attempts
- Health monitoring events (not responding, error dialogs, memory limit breaches, CPU-hung detection, title changes)
- Group and application add/edit/delete operations
- Errors and warnings
- Startup terminations



Log Format

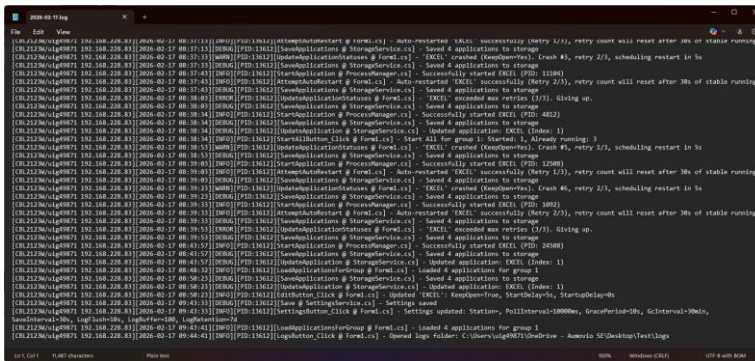
[HOSTNAME/USER IP][2025-06-15 14:30:00][INFO][PID:12345][StartApplication @ ProcessManager.cs] - Successfully started notepad (PID: 9876)

Logs



Log Retention

Logs older than the configured retention period (default: 7 days) are automatically deleted. You can change this in **Settings ? Log Retention (days)**.





Security

Complete Application User Guide

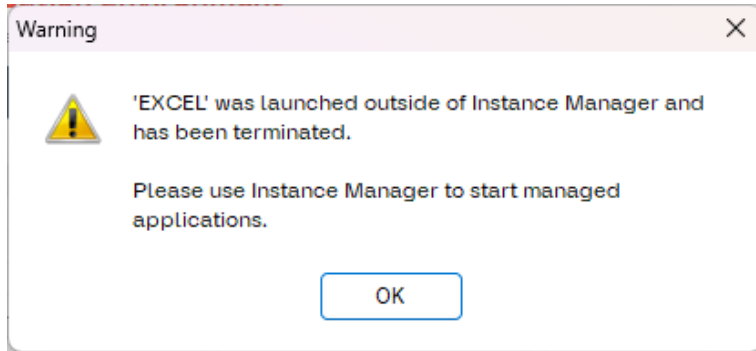
February 20, 2026

IT/LMA – John Moises Paunlagui

Enabling Function

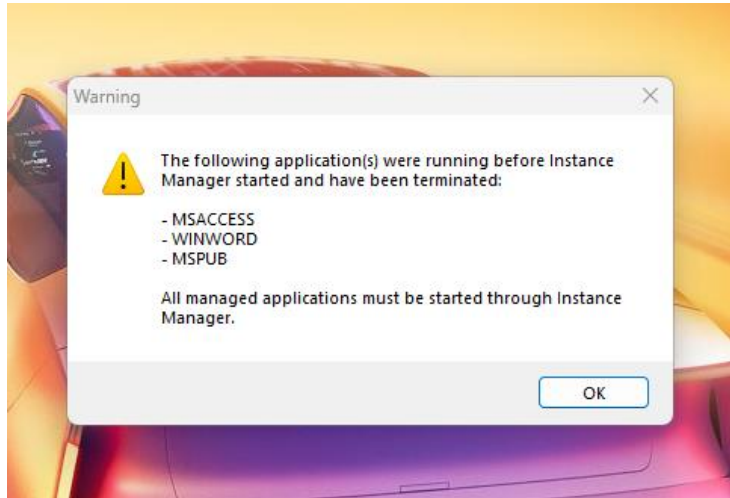
Security

Managed applications can **only** be launched through its interface.



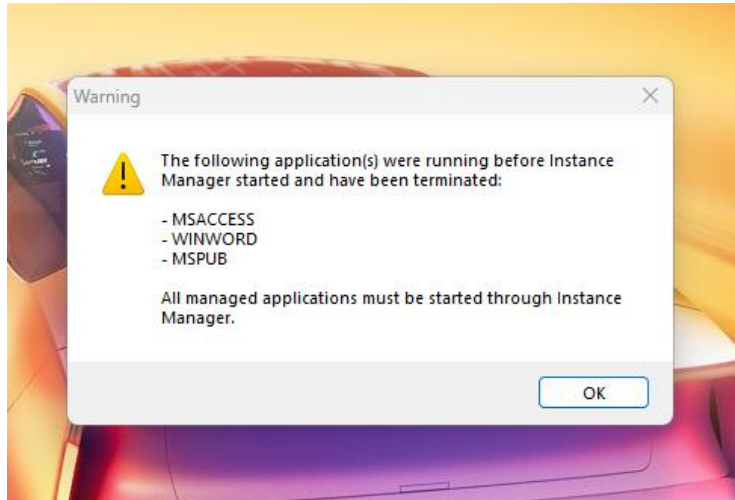
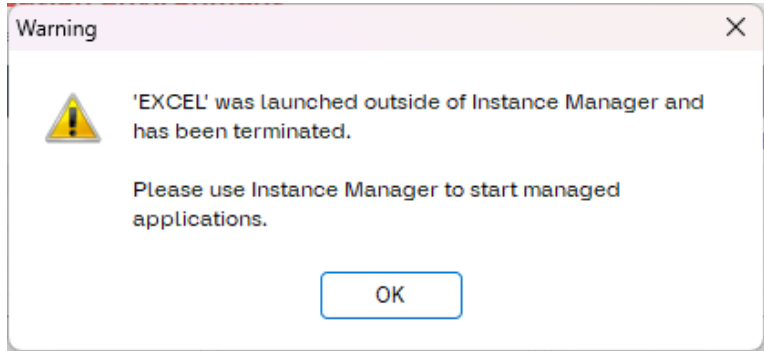
How It Works

- When you start an app through Instance Manager, it's added to an authorized list
- At each poll interval (default: 10 seconds, configurable in Settings), the watchdog checks all managed applications
- If a managed app is running but **not** in the authorized list, it was launched externally
- The unauthorized process is **killed immediately**
- A warning notification is shown (once per attempt)



Security

Managed applications can **only** be launched through its interface.

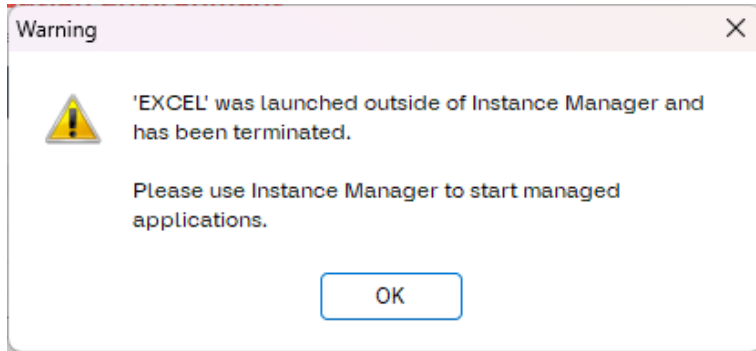


Scenarios

Scenario	What happens
User starts app via Instance Manager	Allowed – added to authorized list
User double-clicks the .exe directly	Killed within one poll interval + warning shown
User opens app via shortcut	Killed within one poll interval + warning shown
App was running before Instance Manager started	Terminated on startup + notification
App crashes and auto-restarts (Keep Open)	Allowed – re-authorized automatically
Same app running in another group	Shown as "Running (Other Group)" – must stop in the other group first

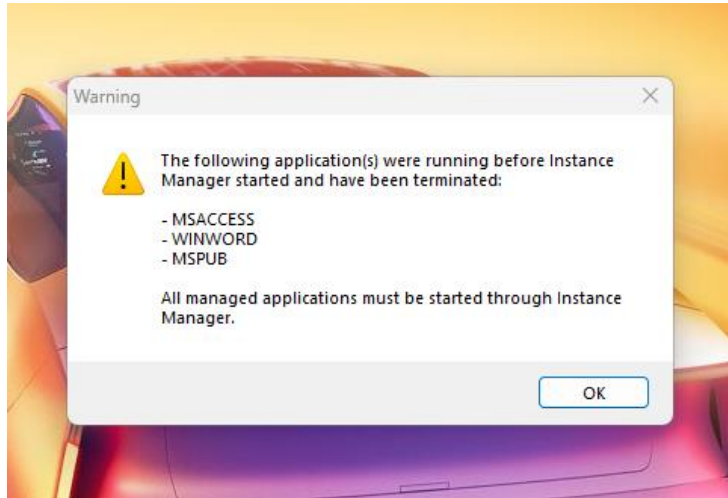
Security

Managed applications can **only** be launched through its interface.



Why this matters

- Ensures only one instance of each managed application runs
- Provides an audit trail of who started what and when
- Prevents circumventing instance management by launching apps directly



FAQs

Complete Application User Guide

Frequently Asked Questions

Can I manage applications that require Administrator privileges?

- No. Instance Manager runs at standard user level and cannot manage elevated processes. If your managed application requires elevation, Instance Manager won't be able to start or stop it.

What happens if I close Instance Manager while applications are running?

- The managed applications **continue running** – Instance Manager does not stop them on exit. However, the watchdog will no longer monitor them. When you restart Instance Manager, any still-running managed apps will be terminated.

Can I manage the same application in multiple groups?

- Yes, you can add the same executable to different groups. Each entry is tracked independently. However, you cannot run the same application simultaneously from multiple groups. If you try to start an application that is already running in another group, Instance Manager will show a warning telling you which group its running in. You must stop it in the other group first before starting it in a new one. The status column will show Running (Other Group) in dark cyan for entries whose executable is running from a different group.

Frequently Asked Questions

Does Instance Manager modify my applications?

- No. Instance Manager only starts and stops processes. It never modifies any application files.

What if my application doesn't have a visible window?

- Instance Manager detects running applications by checking for a main window handle. Applications that run purely in the background (no window at all) may not be detected as "Running" and could be falsely treated as crashed.

What if my application shows an error dialog (unhandled exception)?

- If a Keep Open application becomes unresponsive due to an error dialog (e.g., an unhandled .NET exception dialog), Instance Manager will detect the unresponsive state after two consecutive poll cycles and force-kill the process. This allows auto-restart to take over without requiring manual intervention.

Can I change the watchdog polling interval?

- Yes. Click **Settings** and adjust the **Status Poll Interval** value. The default is 10000ms (10 seconds). Lower values make detection faster but use more CPU.

Frequently Asked Questions

What if my application shows an error dialog (unhandled exception)?

Instance Manager has multiple layers of detection for error dialogs:

1. **Error Dialog Pattern Matching** – If the dialog's window title contains a known error keyword (e.g., "unhandled exception", "NullReferenceException", "fatal error"), it is detected automatically and the process is force-killed after 2 consecutive poll cycles.
2. **Not Responding Detection** – If the error dialog blocks the UI thread and the window becomes unresponsive, it is detected via `Process.Responding`` and force-killed after the configured timeout.
3. **Title Change Detection** – If the error dialog's title doesn't match any known pattern (e.g., a WinForms `ThreadExceptionDialog`` that uses the app's product name as its title), enable **Detect Title Change** in the Edit dialog. The watchdog will detect the title change and force-kill the process.

All three mechanisms trigger auto-restart for Keep Open applications.

Frequently Asked Questions

How do I set a memory limit for my application?

1. Click **Edit** on the application
2. Set **Memory Limit (MB)** to your desired maximum (e.g., 500 for 500 MB)
3. Click OK
4. If the application's working set exceeds this limit, it will be force-killed and auto-restarted (if Keep Open is enabled)

Tip: Monitor your app's normal memory usage in Task Manager first, then set the limit to 2–3× the expected peak.

What is the "Last Exit Code" in the Edit dialog?

The Last Exit Code shows the exit code from the most recent process termination. A value of 0 typically means the process exited normally. A non-zero value (shown in red with "abnormal") usually indicates a crash or error.

Common non-zero codes:

- 1073741819** (0xC0000005) – Access violation
- 1073740791** (0xC0000409) – Stack buffer overrun
- 532462766** (0xE0434352) – .NET unhandled exception

Frequently Asked Questions

Where is my data stored?

All data files are in the same folder as **IntelligentMutexExecutionEnvironment.exe** :

- `applications.json` – your managed applications
- `groups.json` – your groups
- `settings.json` – station name, performance, and logging configuration
- `logs/` – daily log files

How do I reset everything?

Delete `applications.json`, `groups.json`, and `settings.json`, then restart Instance Manager.

Troubleshooting

Complete Application User Guide

Troubleshooting Guide

Application Won't Start

1. Check that the file path is correct – click **Edit** and verify the Directory
2. Make sure the `.exe` file exists at that path
3. Check the log file for error messages
4. Verify you have permission to run the application

Application Won't Stop

1. Check the log file for "Access denied" errors
2. The application may have already closed externally
3. Click **Refresh** to force a status update
4. Try stopping it via Task Manager as a last resort

Status Stuck on "Starting..."

1. The application has been launched but its window hasn't appeared yet. Possible causes:
2. The application is still loading (wait for the grace period, default: 10 seconds, configurable in Settings)
3. The application runs without a visible window (not supported for status detection)
4. The application crashed immediately after launch (check logs)

Troubleshooting Guide

Status Shows "Failed"

The application has exhausted its maximum retry attempts. To fix:

1. Investigate why the application keeps crashing
2. Click **Edit** on the application
3. Reset the **Retry Count** to 0
4. Click OK
5. Click **Start** to try again

Application Keeps Getting Killed by Memory Limit

The memory limit may be set too low for the application's normal operation:

1. Click **Edit** on the application
2. Either increase the **Memory Limit (MB)** value or set it to `0` to disable
3. Click OK

Tip: Use Task Manager to observe the application's working set under normal conditions and set the limit well above that baseline.

Troubleshooting Guide

Application Keeps Getting Killed by Title Change Detection

The application legitimately changes its window title (e.g., showing document names, status text, or progress):

1. Click **Edit** on the application
2. Uncheck **Detect Title Change**
3. Click OK

Application Keeps Getting Killed by CPU-Hung Detection

1. The application legitimately uses sustained high CPU (e.g., rendering, video encoding):

CPU-hung detection is always active and cannot be disabled per-app. However, it requires **95% CPU usage across all cores for 3 consecutive poll cycles** (default: 30 seconds total). Most legitimate high-CPU operations won't trigger this unless they consume 100% of all CPU cores for an extended period.

2. If this is a problem, increase the **Status Poll Interval** in Settings to give the detection more time between checks.

Troubleshooting Guide

Unauthorized Launch Warning Keeps Appearing

This means someone (or something) keeps trying to launch the managed application outside of Instance Manager. Check for:

1. Scheduled tasks that launch the application
2. Startup items in Windows
3. Other users launching the application directly

Troubleshooting Guide

Logs Not Being Created

1. Make sure Instance Manager has write permission in its directory
2. Check if the `logs/` folder exists – if not, try creating it manually
3. Restart Instance Manager

Data File Corruption

If `applications.json` or `groups.json` becomes corrupted:

1. Check for a `.bak` or `.tmp` backup file in the same directory
2. Rename the `.tmp` file to replace the corrupted file
3. If no backup exists, delete the corrupted file (you'll lose that data)
4. Restart Instance Manager

System Requirements

Complete Application User Guide

System Requirement

Minimum Requirement

Requirement	Minimum
Operating System	Windows XP SP3 or later
.NET Framework	4.0 or higher
RAM	50 MB available
Disk Space	5 MB + log files
Display	1024×768 or higher

Thank you